



Unsupervised Concept Discovery and Evaluation

for Medical Vision-Language Models

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Presenter: Emmanuel.

Good morning everyone. Today we present our work on Unsupervised Concept Discovery and Evaluation for Medical Vision-Language Models, focusing on extracting interpretable features from chest X-ray embeddings.



Introduction

Why Are Medical VLMs Opaque?

- Medical **Vision-Language Models** (VLMs) encode images and text in a shared representation
 - Strong performance across clinical tasks (classification, retrieval, report generation)
- However, their latent representations are **high-dimensional** and **polysemantic**
 - Individual dimensions mix multiple clinical meanings
 - Difficult to establish what each dimension represents
- This limits **interpretability** and **trustworthiness** in safety-critical applications
- **Concept-based explanations** aim to express these representations through recognizable anatomy and clinical findings

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Unsupervised Concept Discovery and Evaluation

Introduction

Good morning everyone. Today we present our work on Unsupervised Concept Discovery in Medical Vision-Language Models. Medical VLMs like BiomedCLIP are very powerful: they encode both images and text in a shared latent space and achieve strong clinical performance. However, these representations are high-dimensional and polysemantic, meaning that individual latent dimensions mix multiple clinical meanings. This makes it very hard to understand *what* the model has actually learned, which is a problem in safety-critical medical applications where trust and accountability are essential. Concept-based explanations try to solve this by mapping the representation onto recognizable clinical concepts.

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- Medical **Vision-Language Models** (VLMs) encode images and text in a shared representation
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Introduction

Our Approach

- Following **MedConcept**, we study whether SAE features extracted **without diagnostic supervision** can be named and assessed against radiology reports
- We **adapt** the original pipeline:
 - From 3D abdominal CT → **2D chest radiographs**
 - Backbone: **BiomedCLIP** (frozen ViT-B/16)
 - New dedicated concept vocabulary and activation selection
- **Objective:** Assess what the available evidence supports about interpretability
 - *Not* to develop a diagnostic classifier

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Unsupervised Concept Discovery and Evaluation

Introduction

Our work builds on the MedConcept framework. The key question we ask is: can Sparse Autoencoder features, extracted in a completely unsupervised way without any disease labels, be meaningfully named and then validated against actual radiology reports? We adapt the original pipeline, which was designed for 3D abdominal CT scans, to work with 2D chest radiographs. We replace the backbone with BiomedCLIP and build a new dedicated vocabulary of clinical concepts. It is important to note that our objective is not to build a diagnostic classifier. Instead, we want to assess what the available evidence actually supports about the interpretability of these discovered features.

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Introduction

Key Contributions & Findings

Contributions:

1. An implemented pipeline with **phrase-level semantic grounding**
2. Analysis of concept groups at **multiple clustering resolutions**
3. An independent **multi-evaluator audit** of the LLM judge

Key Findings:

- Report-based alignment is **low on average** (7.21% Aligned) but **highly uneven** across clinical subdomains (up to 49.23% in specific clusters)
- The **choice of judge itself** is a major source of disagreement: MedGemma agrees with human consensus only 56% of the time vs. 68–76% for commercial LLMs

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Unsupervised Concept Discovery and Evaluation

Introduction

Our three main contributions are: first, a fully implemented pipeline that grounds SAE features to clinical phrases; second, an analysis of how alignment varies across groups of concepts at different clustering resolutions; and third, an independent audit of the LLM judge using both human reviewers and multiple commercial models. Our key findings are that report-based alignment is only 7.2 percent on average, but this average is misleading because it is highly uneven: some concept clusters reach nearly 50 percent alignment. Furthermore, we found that the choice of the evaluator model itself is a major source of disagreement. MedGemma agreed with our human consensus only 56 percent of the time, while commercial LLMs reached up to 76 percent. This tells us that the judge, not just the concepts, is a critical variable in measuring interpretability.

Introduction Key Contributions & Findings

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Related Work

From Predefined Concepts to Discovery

- **Supervised concept methods** offer semantic interfaces but require predefined annotations:
 - **TCAV**: Sensitivity along user-defined concept directions
 - **Concept Bottleneck Models**: Predict annotated concepts as an intermediate representation
- **Unsupervised discovery** removes this requirement:
 - **Sparse dictionary learning**: Decomposes polysemantic activations into monosemantic features without predefined concepts
 - **Olah et al.**: Interprets resulting units as *circuits* rather than opaque activations
- Recent evolutions:
 - **Discover-then-Name**: SAE discovery → naming → concept bottlenecks
 - **SpLiCE**: Decomposes CLIP embeddings directly into sparse combinations of existing textual concepts

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Related Work

Let's look at the related work. Traditional interpretability approaches fall into two camps. On one side, supervised methods like TCAV and Concept Bottleneck Models give us nice semantic interfaces, but they require manually defined and annotated concept sets upfront. On the other side, unsupervised methods based on sparse dictionary learning remove this requirement entirely. They decompose polysemantic activations, where a single neuron responds to many concepts, into monosemantic features. Olah et al. go further and interpret these features and their connections as interpretable circuits. Building on these ideas, Discover-then-Name first discovers features with an SAE and then names them, while SpLiCE takes a different approach and directly decomposes CLIP embeddings into sparse combinations of existing textual concepts without any new representation learning.

Related Work

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Related Work

VLMs and Medical Grounding

- **Pach et al.:** Evaluate SAE features in VLMs using a human-study benchmark and interventions — going beyond reconstruction alone
- **Gong et al.:** Dissect neuron activations in medical diagnostic models to build concept-level explanations directly from opaque networks
- **MedConcept** — the framework we build upon:
 - Sparse decomposition of *Merlin* CT embeddings
 - Ontology-derived vocabulary + cosine-based naming
 - Frozen MedGemma judgments of concept-report consistency
- **Our adaptation:** We retain the discovery/evaluation separation, while changing:
 - Image domain (3D CT → 2D chest X-ray)
 - Backbone (Merlin → BiomedCLIP)
 - Vocabulary construction and activation selection

Unsupervised Concept Discovery and Evaluation

Related Work

In the medical domain specifically, Pach et al. evaluated SAE features in vision-language models using human studies and interventions, going beyond just measuring reconstruction quality. Gong et al. dissected neuron activations in medical imaging models to build interpretable explanations, motivating a neuron-to-concept alignment strategy. The framework we directly build upon is MedConcept, which combines sparse decomposition of 3D CT embeddings with an ontology-derived vocabulary and MedGemma-based evaluation. We retain MedConcept's separation between discovery and evaluation, but we change the image domain from 3D CT to 2D chest X-rays, swap the backbone from Merlin to BiomedCLIP, and build a new vocabulary with a different activation selection strategy.

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Related Work

Evaluating Explanations

- Different evaluation methods capture **distinct aspects** of explanation quality:
 - **Reconstruction**: How well are embeddings preserved?
 - **Human inspection**: Are features semantically coherent?
 - **Interventions**: Do they affect model behavior?
 - **Report entailment**: Is there textual support?
- These approaches provide **complementary** evidence — none is sufficient alone
- **LLM-as-a-judge** scales well but introduces vulnerabilities
 - Position bias and sensitivity to prompt format
- Our approach: **label permutation + independent multi-evaluator validation**

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Unsupervised Concept Discovery and Evaluation

Related Work

Evaluating the quality of explanations is an open challenge, because different evaluation methods capture different things. Reconstruction tells us if embeddings are preserved, human inspection checks semantic coherence, interventions test whether features actually affect behavior, and report entailment checks textual support. These are complementary: none alone is sufficient. Using an LLM as a judge scales better than manual review, but the literature has shown it is susceptible to position bias: the order in which answer options are presented can affect the verdict. To address this, we use label permutation across three forward passes to reduce letter bias, and we validate our judge's reliability with an independent human audit and comparison against multiple commercial LLMs.

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Research Gaps

Four Gaps Guide Our Analysis

1. Reconstruction $\neq$ Correct Naming

SAE sparsity and fidelity do not guarantee correct labels — *vocabulary coverage* and *image-text geometry* constrain naming

2. Dependence on External Resources

Every stage from raw activations to a clinical claim relies on external tools (ontology, generative phrasing, LLM judges) — biases can be mistaken for properties of the representation

3. Reports as Selective Evidence

A report is partial: unmentioned concepts may be visible, absent, or incorrectly named — judges can confuse *omission* with *contradiction*

4. Independent Labels Omit Relationships

Assigning labels independently ignores anatomical and causal relationships between findings — embedding similarity $\neq$ anatomical hierarchy

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Research Gaps

We identified four research gaps that guide our entire analysis. First, achieving good reconstruction does not mean the nearest text label is correct. The vocabulary itself and the geometry of the image-text embedding space constrain the naming process. Second, the pipeline depends on external resources at every stage: the ontology vocabulary, a generative model for phrase expansion, and one or more LLMs as judges. Biases or errors in any of these components can be mistaken for properties of the learned representation itself. Third, radiology reports are selective evidence. Radiologists prioritize acute findings and often omit normal anatomy. An automated judge can confuse this omission with contradiction, unfairly penalizing correct concepts. Fourth, assigning labels independently to each feature ignores the relationships between findings. Embedding similarity does not establish an anatomical hierarchy or a causal relation.

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Research Gaps

Our Response

These gaps point to the broader difficulty of **assessing interpretability in a scalable and reproducible way**. We address them by:

- **Hierarchical clustering** at multiple resolutions to examine whether semantic groups produce useful concept categories
 - Recognizing that embedding similarity alone does not validate clinical groupings
- **Human + multi-model audit** to specifically address **evaluator reliability**
 - 3 human annotators + 3 commercial LLMs on 25 stratified pairs
 - Surfaces the pipeline's dependence on the judge's quality
- **Label permutation** across forward passes to mitigate position bias in the LLM judge

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Together, these four gaps point to a broader problem: assessing interpretability in a way that is both scalable and reproducible is fundamentally difficult. We address these gaps in three concrete ways. First, we apply hierarchical clustering at multiple resolutions to examine whether semantic groupings of concepts produce useful categories. However, we acknowledge that embedding similarity alone does not validate clinical groupings. Second, and most importantly, we run a human plus multi-model audit specifically targeting evaluator reliability. Three team members independently evaluated 25 concept-report pairs, and we compared their consensus against MedGemma and three commercial LLMs. This directly surfaces how much the pipeline's results depend on the judge's quality. Third, we mitigate position bias in the LLM judge by permuting the label-to-letter mapping across three forward passes and averaging the probabilities.



Methodology

Datasets and Feature Extraction

- **NIH Chest X-ray 10k:** 7,500 images used exclusively to train the Sparse Autoencoder (SAE), and 2,500 for validation[cite: 3]. No text or labels were used[cite: 3].
- **Open-I:** 3,666 image-report pairs used for semantic evaluation[cite: 3]. Reports concatenate Findings and Impression sections[cite: 3].
- **Feature Extraction:** BiomedCLIP's frozen ViT-B/16 image encoder produced the visual embeddings for both datasets[cite: 3].

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Methodology

Let's discuss our methodology, starting with the datasets[cite: 3]. We kept discovery and evaluation strictly separate. We trained the Sparse Autoencoder on 7,500 images from the NIH dataset without any text or labels[cite: 3]. For evaluation, we used the Open-I dataset, which provides paired images and radiology reports[cite: 3]. All visual embeddings were extracted using BiomedCLIP's frozen ViT-B/16 encoder[cite: 3].

Methodology
Datasets and Feature Extraction

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- **Concept Selection:** 100 clinical parent concepts extracted and filtered from Open-I Problems metadata[cite: 3].
- **Expansion:** Concepts were mapped via UMLS to SNOMED CT and MeSH to retrieve synonyms and Unique Identifiers (CUIs)[cite: 3].
- **Phrase Generation:** Gemini 3.1 Flash-Lite generated 5 descriptive phrases per concept[cite: 3].
- **Encoding:** Phrases were encoded with BiomedCLIP's PubMedBERT into a 998×512 matrix[cite: 3].

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To name the features learned by the SAE, we built a dictionary of medical concepts[cite: 3]. We started with 100 parent concepts from Open-I metadata and linked them to standardized vocabularies like SNOMED CT using UMLS[cite: 3]. To get radiology-style descriptions, we used Gemini 3.1 Flash-Lite to generate 5 short phrases per concept[cite: 3]. These phrases were then encoded with PubMedBERT to map them into the same latent space as the images[cite: 3].



Methodology

Sparse Autoencoder and Concept Assignment

- **SAE Architecture:** Input dimension $d = 512$, hidden dimension $m = 1024$ (overcomplete factor of 2)[cite: 3].
- **Training:** Optimization includes an L_1 penalty to encourage sparse activations[cite: 3].
- **Naming:** Each learned feature direction (decoder column) is assigned a phrase based on the highest cosine similarity[cite: 3].
- **Inference:** For a new image, active neurons contributing to $> 90\%$ of the total positive activation are selected to form the predicted concept set[cite: 3].

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Methodology

Our Sparse Autoencoder maps the 512-dimensional embeddings into a 1,024-dimensional hidden space[cite: 3]. By applying an L1 penalty, we force the model to use only a few active units per image[cite: 3]. We assigned names to these 1,024 features by finding the closest phrase in our dictionary using cosine similarity[cite: 3]. During inference, we select the top features that account for 90 percent of the positive activation for a given image[cite: 3].

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Sparse Autoencoder and Concept Assignment

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Methodology

Semantic Evaluation and Validation

- **Evaluator:** Frozen MedGemma 1.5 4B[cite: 3].
- **Verdicts:**
 - **Aligned:** Report explicitly supports the concept[cite: 3].
 - **Unaligned:** Report explicitly negates/contradicts the concept[cite: 3].
 - **Uncertain:** Concept is unmentioned or ambiguous[cite: 3].
- **Bias Mitigation:** 3 forward passes with permuted label-to-letter mappings to reduce position bias[cite: 3].

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Methodology

To evaluate if the extracted concepts match the radiology reports, we used MedGemma 1.5 4B as a judge[cite: 3]. The model classifies each concept-report pair as Aligned, Unaligned, or Uncertain[cite: 3]. To prevent the LLM from simply favoring the first option, we rotated the answer choices across three separate passes and averaged the probabilities[cite: 3].

Methodology

Semantic Evaluation and Validation

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Experiments

Sparse Reconstruction and Grounding

- **Reconstruction Fidelity:** Mean Squared Error (MSE) of 3.32×10^{-5} and mean cosine similarity of 0.9916 on the validation set[cite: 3].
- **Sparsity:** Average of 44.0 active units out of 1024 ($\sim 4.3\%$ L_0 sparsity) per image[cite: 3].
- **Takeaway:** The SAE achieves near-lossless visual reconstruction under a strict sparse activation regime[cite: 3]. However, fidelity does not guarantee semantic monosemanticity[cite: 3].

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Experiments

Moving to our results, the SAE training was highly successful in terms of reconstruction[cite: 3]. We achieved a very low Mean Squared Error and a cosine similarity above 0.99 on the validation set[cite: 3]. It did this while maintaining high sparsity, using an average of only 44 active units out of 1024 per image[cite: 3]. However, as we will see, this mathematical fidelity does not automatically mean the features are perfectly interpretable clinically[cite: 3].

Experiments

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Experiments

Report Alignment and Cohort Analysis

- **Cohort Performance** ($N = 500$):
 - **Uncertain:** 70.30%[\[cite: 3\]](#)
 - **Unaligned:** 22.49%[\[cite: 3\]](#)
 - **Aligned:** 7.21%[\[cite: 3\]](#)
- **Omission Bias:** The dominance of Uncertain verdicts is due to radiologists omitting normal anatomy and unmentioned concepts[\[cite: 3\]](#).
- **Unaligned Rate:** Captures both false-positive concept predictions and explicit clinical negations[\[cite: 3\]](#).

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Experiments

When we tested the extracted concepts against actual reports on a 500-image cohort, the alignment was surprisingly low, averaging about 7.2 percent[\[cite: 3\]](#). Over 70 percent of concepts were marked as Uncertain[\[cite: 3\]](#). This is heavily driven by reporting omission bias: radiologists document acute pathology but skip normal anatomy, which the SAE still extracts[\[cite: 3\]](#). The 22.5 percent Unaligned rate includes both model false-positives and explicit negations in the text[\[cite: 3\]](#).

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Experiments

Hierarchical Concept Clustering

- **Approach:** Agglomerative hierarchical clustering on the 100 parent concept embeddings using cosine distance[cite: 3].
- **Localized Alignment:**
 - Focal, specific clusters score highly: *Costophrenic Angle / Sulcus* achieves **49.23%** Aligned[cite: 3].
 - Broad groups score poorly: *Mixed Cardiopulmonary* yields only 8.62% Aligned[cite: 3].
- **Conclusion:** Clinical agreement is highly localized and unevenly distributed across subdomains[cite: 3].

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Experiments

To see if alignment varied by topic, we clustered the 100 parent concepts[cite: 3]. We found that clinical agreement is highly localized[cite: 3]. For example, focal anatomical findings like the "Costophrenic Angle" achieved over 49 percent alignment[cite: 3]. Conversely, broad or mixed clusters performed very poorly[cite: 3]. This shows that the SAE learns certain specific features much better than general ones[cite: 3].

Experiments

Hierarchical Concept Clustering

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Experiments

Evaluator Audit and Qualitative Case Studies

- **Human Consensus:** 3 human experts audited 25 concept-report pairs[cite: 3].
- **Model Agreement vs. Humans:**
 - **MedGemma 1.5 4B:** 56.0% agreement[cite: 3].
 - **Commercial LLMs** (GPT-5.6, Claude 5, Gemini 3.1 Pro): 68.0% — 76.0% agreement[cite: 3].
- **Failure Modes:** MedGemma suffered from negation parsing errors (e.g., misclassifying "no pleural effusion" as an Aligned "large effusion")[cite: 3].

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Experiments

Finally, we audited our judge model[cite: 3]. Three experts manually evaluated 25 concept-report pairs[cite: 3]. We discovered that MedGemma agreed with human consensus only 56 percent of the time, while commercial models like Gemini 3.1 Pro reached up to 76 percent[cite: 3]. A major failure mode for MedGemma was negation: it would read "no pleural effusion" and incorrectly mark the concept "large effusion" as Aligned[cite: 3]. This proves that automated judges cannot yet replace expert review[cite: 3].

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Thank you for listening!
Any questions?